

Correlated Beam Misalignment in THz Sensing

PURE Program — research project

Objective

Extend the existing beam-misalignment simulation by replacing independent pointing-error samples with a simple temporally correlated model. The goal is to observe how slowly varying beam misalignment affects sensing performance over several consecutive radar frames. No new probability derivations are required.

Background

In the current model, the pointing error is generated independently for every sensing frame. In practice, platform motion, vibration and beam pointing errors change gradually. Therefore, the current pointing error should depend partly on the previous pointing error.

The beam-misalignment power coefficient is

$$h_k = \exp\left(-\frac{2r_k^2}{w_{\text{mo}}^2}\right), \quad (1)$$

where r_k is the pointing displacement during sensing frame k . Use h_k directly as a received-power coefficient. Do **not** use h_k^2 .

Task 1 — Review the existing simulation

Read the two provided papers and identify:

- how the pointing displacement r is generated;
- how h is calculated;
- where h is included in the radar equation or received signal;
- the meanings of *spot-filling* and *spot-limited* sensing.

Check the simulation code and confirm that

$$P_r = P_{r,\text{aligned}} h. \quad (2)$$

Deliverable. A short list of the relevant equations and code locations.

Task 2 — Generate independent misalignment samples

Generate two independent Gaussian pointing components:

$$x_k \sim \mathcal{N}(0, \sigma^2), \quad y_k \sim \mathcal{N}(0, \sigma^2), \quad (3)$$

and calculate

$$r_k = \sqrt{x_k^2 + y_k^2}. \quad (4)$$

Generate at least 10,000 samples. Plot:

1. x_k versus sample number;
2. r_k versus sample number;
3. a histogram of r_k ;

4. the resulting beam coefficient h_k .

This is the independent (i.i.d.) reference case.

Task 3 — Generate correlated misalignment samples

Replace the independent Gaussian samples with

$$x_k = \rho x_{k-1} + \sqrt{1 - \rho^2} \sigma u_{x,k}, \quad (5)$$

$$y_k = \rho y_{k-1} + \sqrt{1 - \rho^2} \sigma u_{y,k}, \quad (6)$$

where $u_{x,k}, u_{y,k} \sim \mathcal{N}(0, 1)$.

Test $\rho \in \{0, 0.9, 0.99\}$. For each value of ρ , calculate

$$r_k = \sqrt{x_k^2 + y_k^2} \quad \text{and} \quad h_k = \exp\left(-\frac{2r_k^2}{w_{\text{mo}}^2}\right). \quad (7)$$

Plot r_k and h_k versus sensing-frame number, and compare the histograms of r_k obtained for the different values of ρ .

Task 4 — Compare the statistical behaviour

For the i.i.d. case and each correlated case, calculate:

- average value of h_k ;
- minimum value of h_k ;
- percentage of frames for which $h_k < h_{\text{th}}$;
- average number of consecutive frames below the threshold;
- maximum number of consecutive frames below the threshold.

Select one reasonable value of h_{th} , such as 0.1 or 0.2, and use the same threshold in every comparison.

Deliverable. One comparison table.

Task 5 — Add the model to the sensing simulation

Keep the beam coefficient constant during each sensing frame or CPI. For frame k , use

$$P_{r,k} = P_{r,\text{aligned}} h_k. \quad (8)$$

Run the existing range–velocity estimation for a sequence of frames using:

1. aligned sensing;
2. i.i.d. beam misalignment;
3. correlated beam misalignment with $\rho = 0.9$;
4. correlated beam misalignment with $\rho = 0.99$.

Use identical system parameters and noise settings. Record for each frame:

- estimated range;
- estimated velocity;
- whether the correct range–Doppler peak was detected;
- received SNR or received power;
- current value of h_k .

Task 6 — Produce the main results

Prepare the following figures:

1. beam coefficient h_k versus frame number for different ρ ;
2. received SNR versus frame number;
3. correct and incorrect detections versus frame number;
4. number of consecutive failed sensing frames versus ρ .

Also prepare one table containing:

- average range error;
- average velocity error;
- probability of correct detection;
- average consecutive failure duration;
- maximum consecutive failure duration.

Task 7 — Write a short interpretation

Write a brief discussion answering:

1. Does temporal correlation change the average beam loss?
2. Does it change how long poor alignment lasts?
3. Can i.i.d. and correlated misalignment have similar average performance but different failure patterns?
4. Is range or velocity estimation more sensitive?
5. Why may correlated misalignment be more important for tracking than for a single sensing frame?

Scope limits

You are **not** required to:

- derive the probability distribution of the correlated process;
- derive Marcum- Q expressions;
- implement Kalman filtering;
- model compensation delay;
- study beam squint;
- compare OFDM and SC-FDMA;
- perform hardware experiments.

Final submission

The organisation, format and pacing of the submission are left to you. The items below are the expected content; decide yourselves how to package and present them.

- commented MATLAB or Python code;
- the main figures;
- the results table;
- the written interpretation;
- a list of any problems or unclear assumptions encountered.

Recommended references

1. S. Sani, A. Saeed, O. Gurbuz and W. Gerstacker, “Beam Misalignment Fading in THz D2D ISAC: From Communication to Sensing,” IEEE PIMRC, 2025.
2. “OFDM vs. SC-FDMA for Monostatic THz ISAC Sensing Under Beam Misalignment Fading,”

SIU 2026 draft (provided).

3. A. A. Farid and S. Hranilovic, "Outage Capacity Optimization for Free-Space Optical Links With Pointing Errors," *Journal of Lightwave Technology*, 2007.
4. M. S. Bashir and M. R. Bell, "Optical Beam Position Tracking in Free-Space Optical Communication Systems," *IEEE Transactions on Aerospace and Electronic Systems*, 2018.